Exercise #5

Introduction To Artificial Intelligence

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1 A^* vs. DFS

Problem Statement. *True/False: Depth-first search always expands at least as many nodes as A search with an admissible heuristic.*

False. DFS has the best case of $\Theta(d)$ (for the case where there is no backtracking to find a solution). A^* can make no such guarantee, because it ranges by heuristic.

2 $h(n) = 0$ For 8-Puzzle

Problem Statement. *True/False: $h(n) = 0$ is an admissible heuristic for the 8-puzzle.*

True. For positive path costs, $h(n) = 0$ will always be an admissible heuristic.

$$\forall \epsilon \in \mathbb{R}^+, h(n) = 0 \leq C^*(n) = \epsilon$$

3 A^* In Robotics

Problem Statement. *True/False: A is of no use in robotics because percepts, states, and actions are continuous.*

True. A^* operates on discrete states. However, there is always the possibility to create discrete states from continuous states.

4 Breadth-First Search Completeness

Problem Statement. *True/False: Breadth-first search is complete even if zero step costs are allowed.*

True. Per our definition of completeness:

> A search algorithm is complete iff it will find a goal when one exists.

5 Rook Admissibility

Problem Statement. *True/False: Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves.*

False. Using the chess depicted in Figure 1, the Manhattan Distance from $A1$ to $A8$ would be 8; however, a rook could get there in one move.

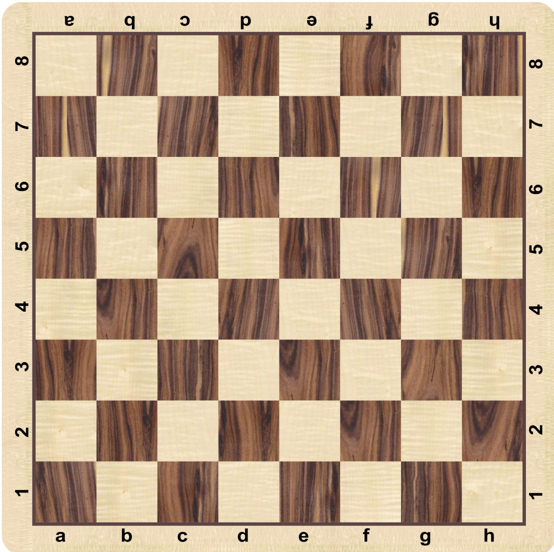

Figure 1: A Standard Chessboard